Creating a Simple WebApp with the Tomcat Server

<u>Step 1:</u> Run the command in order to create a project with the name "**MyMavenWebApp01**"

mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenWebApp01 -DarchetypeArtifactId=maven-archetype-webapp -DinteractiveMode=false

```
vboxuser@niranjan: ~/devops
vboxuser@niranjan:~$ cd devops
vboxuser@niranjan:~/devops$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenWebApp01 -DarchetypeArtifactId=maven-archetype-w
ebapp -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] ------------------< org.apache.maven:standalone-pom >-------------------
[INFO] Building Maven Stub Project (No POM) 1
[INFO] --------------------------------[ pom ]---------------------------------
[INFO]
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO]
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO]
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 2.7 MB/s)
[INFO] ----------------------------------------------------------------------------
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-webapp:1.0
[INFO] ----------------------------------------------------------------------------
[INFO] Parameter: basedir, Value: /home/vboxuser/devops
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenWebApp01
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] project created from Old (1.x) Archetype in dir: /home/vboxuser/devops/MyMavenWebApp01
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  12.788 s
[INFO] Finished at: 2026-04-16T01:02:54+05:30
```

<u>Step 2:</u> Update the pom.xml with the relevant dependencies and the relevant plugins

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

  xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
http://maven.apache.org/maven-v4_0_0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <groupId>com.example</groupId>

  <artifactId>MymavenWebApp01</artifactId>

  <packaging>war</packaging>

  <version>1.0-SNAPSHOT</version>

  <name>MymavenWebApp01 Maven Webapp</name>

  <url>http://maven.apache.org</url>
```

```
 <dependencies>
   <dependency>
     <groupId>junit</groupId>
     <artifactId>junit</artifactId>
     <version>3.8.1</version>
     <scope>test</scope>
   </dependency>
    <!-- Servlet API for Java EE -->
       <dependency>
           <groupId>javax.servlet</groupId>
           <artifactId>javax.servlet-api</artifactId>
           <version>4.0.1</version>
           <scope>provided</scope>
       </dependency>
  </dependencies>
  <build>
  <plugins>
  <plugin>
    <groupId>org.apache.maven.plugins</groupId>
    <artifactId>maven-war-plugin</artifactId>
    <version>3.4.0</version> <!-- Update to latest stable version -->
</plugin>
</plugins>
    <finalName>MymavenWebApp01</finalName>
  </build>
</project>
```

Step 3: Run **mvn clean install** to compile and generate the **War** file

```
vboxuser@niranjan: ~/devops/MyMavenWebApp01
vboxuser@niranjan:~/devops$ cd MyMavenWebApp01/
vboxuser@niranjan:~/devops/MyMavenWebApp01$ mvn clean install
[INFO] Scanning for projects...
[INFO]
[INFO] ----------------------< com.example:MymavenWebApp >-----------------------
[INFO] Building MymavenWebApp Maven Webapp 1.0-SNAPSHOT
[INFO] --------------------------------[ war ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ MymavenWebApp ---
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MymavenWebApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] Copying 0 resource
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ MymavenWebApp ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MymavenWebApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/vboxuser/devops/MyMavenWebApp01/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ MymavenWebApp ---
[INFO] No sources to compile
[INFO]
[INFO] --- maven-surefire-plugin:2.12.4:test (default-test) @ MymavenWebApp ---
[INFO] No tests to run.
[INFO]
[INFO] --- maven-war-plugin:3.4.0:war (default-war) @ MymavenWebApp ---
[INFO] Packaging webapp
[INFO] Assembling webapp [MymavenWebApp] in [/home/vboxuser/devops/MyMavenWebApp01/target/MyMavenWebApp]
[INFO] Processing war project
[INFO] Copying webapp resources [/home/vboxuser/devops/MyMavenWebApp01/src/main/webapp]
[INFO] Building war: /home/vboxuser/devops/MyMavenWebApp01/target/MyMavenWebApp.war
```

```
[INFO]
[INFO] --- maven-install-plugin:2.4:install (default-install) @ MymavenWebApp ---
[INFO] Installing /home/vboxuser/devops/MyMavenWebApp01/target/MyMavenWebApp.war to /home/vboxuser/.m2/repository/com/example/MymavenWebApp/1
.0-SNAPSHOT/MymavenWebApp-1.0-SNAPSHOT.war
[INFO] Installing /home/vboxuser/devops/MyMavenWebApp01/pom.xml to /home/vboxuser/.m2/repository/com/example/MymavenWebApp/1.0-SNAPSHOT/Mymav
enWebApp-1.0-SNAPSHOT.pom
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  5.795 s
[INFO] Finished at: 2026-04-16T01:06:22+05:30
[INFO] ------------------------------------------------------------------------
vboxuser@niranjan:~/devops/MyMavenWebApp01$
```

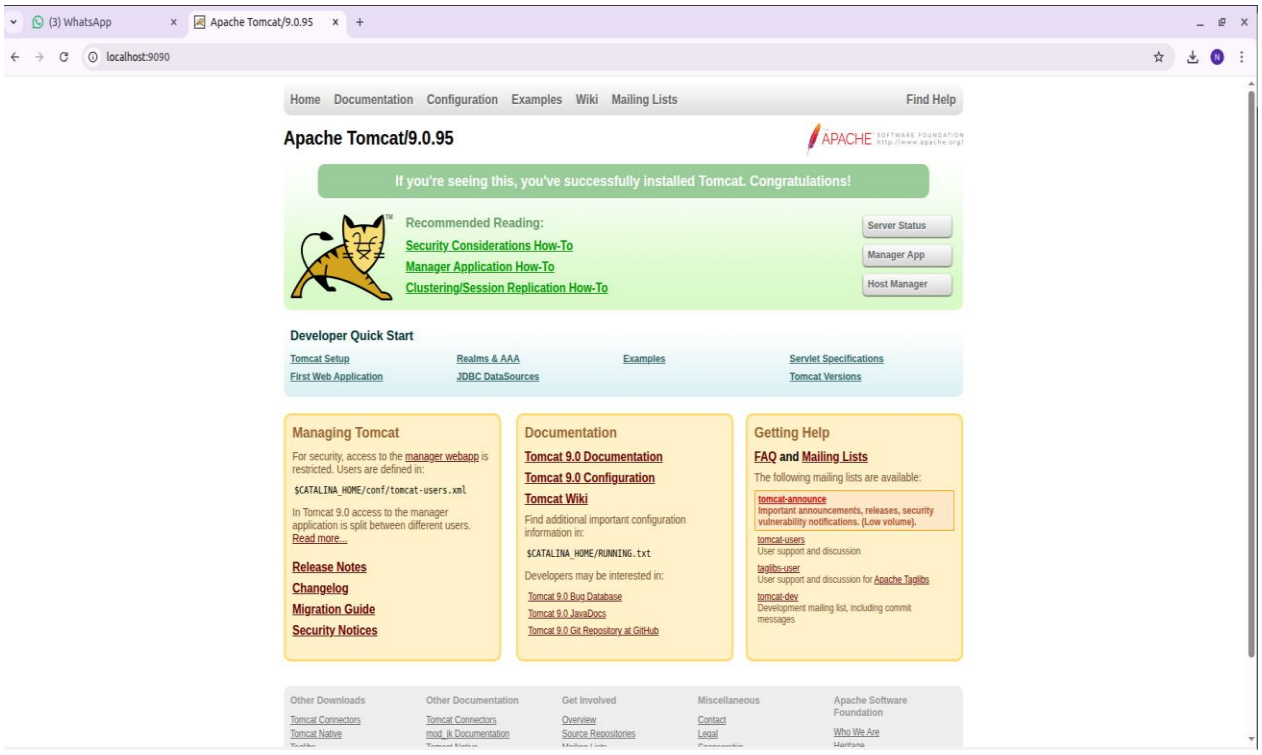
Step 4: Run **tree** in order to check if the War file exists

```
vboxuser@niranjan: ~/devops/MyMavenWebApp01
vboxuser@niranjan:~/devops/MyMavenWebApp01$ tree
.
├── pom.xml
├── src
│   └── main
│       ├── resources
│       └── webapp
│           ├── index.jsp
│           └── WEB-INF
│               └── web.xml
└── target
    ├── classes
    ├── maven-archiver
    │   └── pom.properties
    ├── MyMavenWebApp
    │   ├── index.jsp
    │   ├── META-INF
    │   └── WEB-INF
    │       ├── classes
    │       └── web.xml
    └── MyMavenWebApp.war

12 directories, 7 files
vboxuser@niranjan:~/devops/MyMavenWebApp01$
```

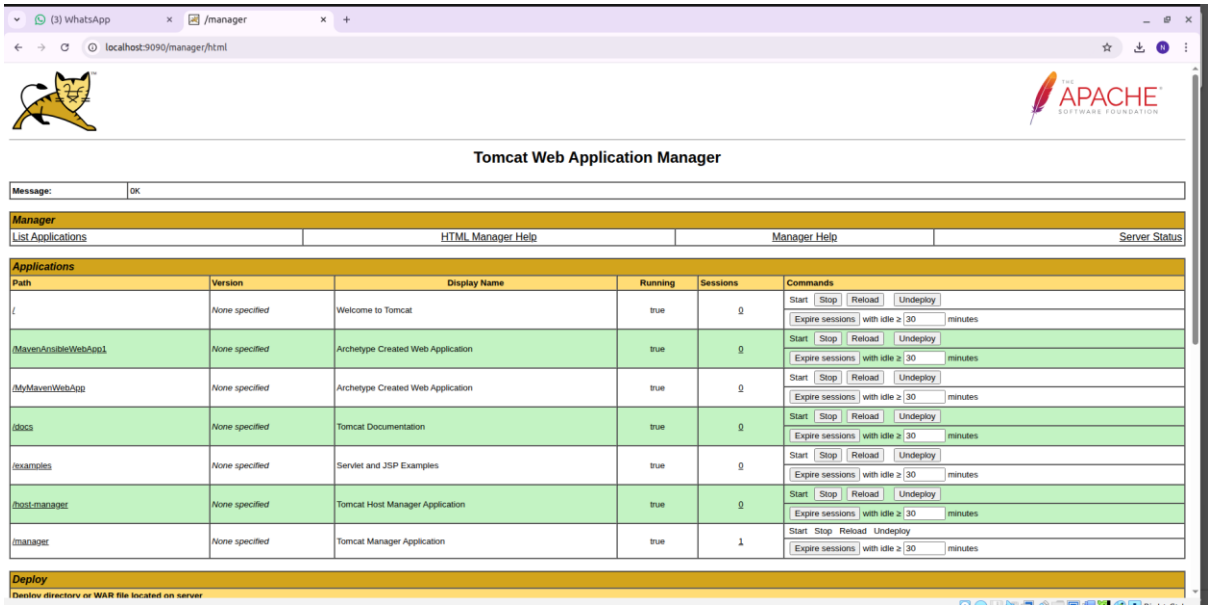
Step 5: change dir to opt/tomcat/bin and run the command **./startup.sh** to start the tomcat server.

```
root@niranjan: /opt/tomcat/bin
vboxuser@niranjan:~$ sudo -i
[sudo] password for vboxuser:
root@niranjan:~# cd /
root@niranjan:/# cd opt/tomcat/bin
root@niranjan:/opt/tomcat/bin# ./startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
root@niranjan:/opt/tomcat/bin#
```

Step 6: now we can see the tomcat server that is started.

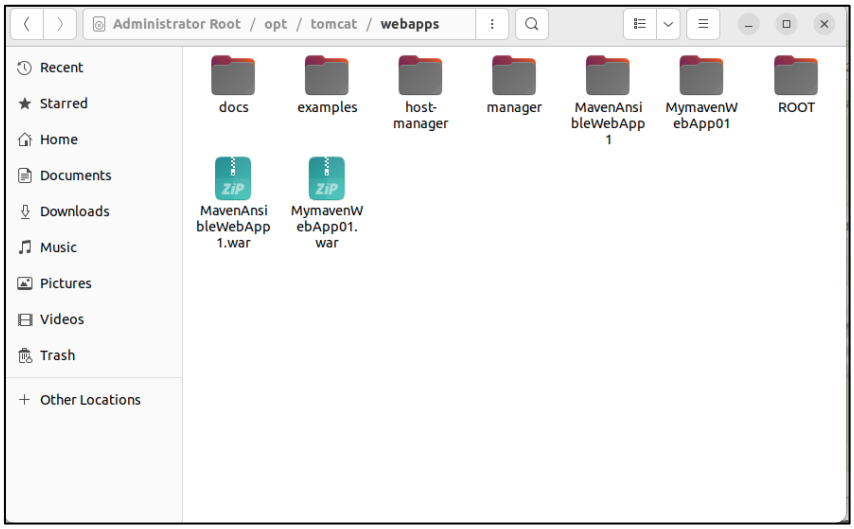


Log in to Manager App using username and password

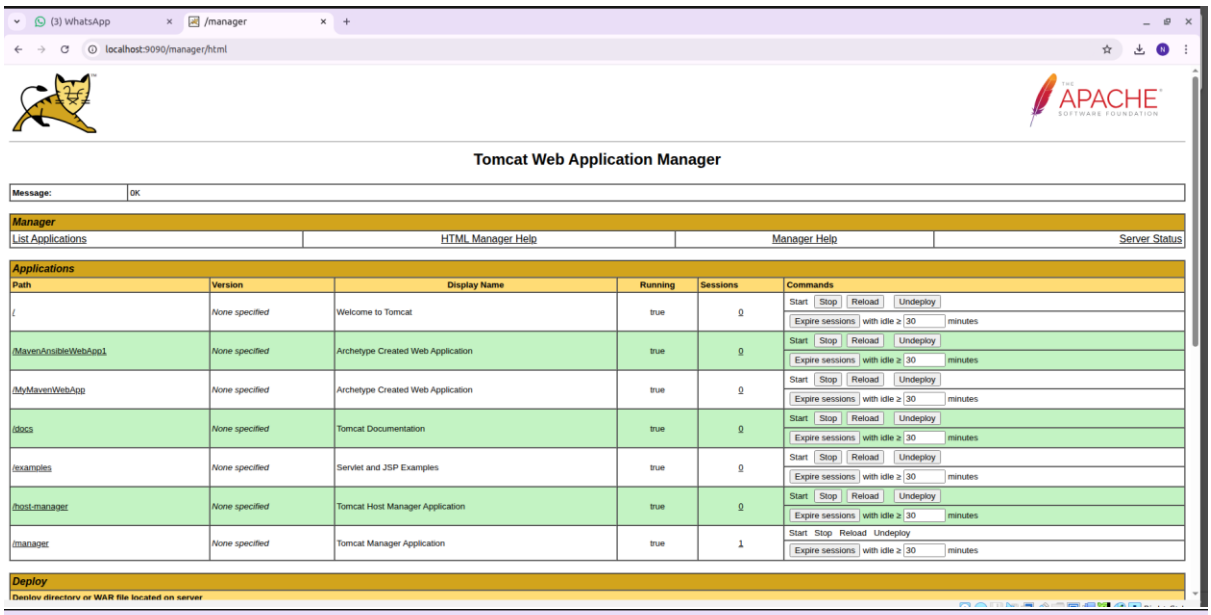


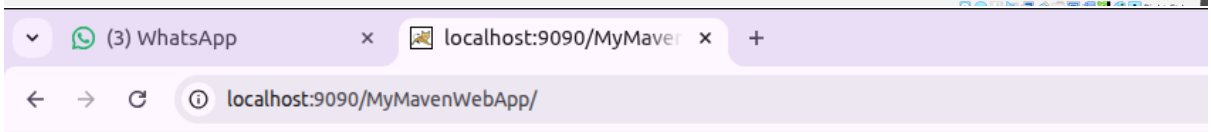
Step 7: Before Running the Web application on tomcat Server change the **index.jsp** file

Step 8: Now before running it, we need to copy the War file created and paste in the file path of tomcat file i.e**. /opt/tomcat/webapps.**After Pasting we can run the tomcat server easily



<u>Step 9:</u> Now we can run the tomcat server with the filename **MyMavenWebApp** to get the output

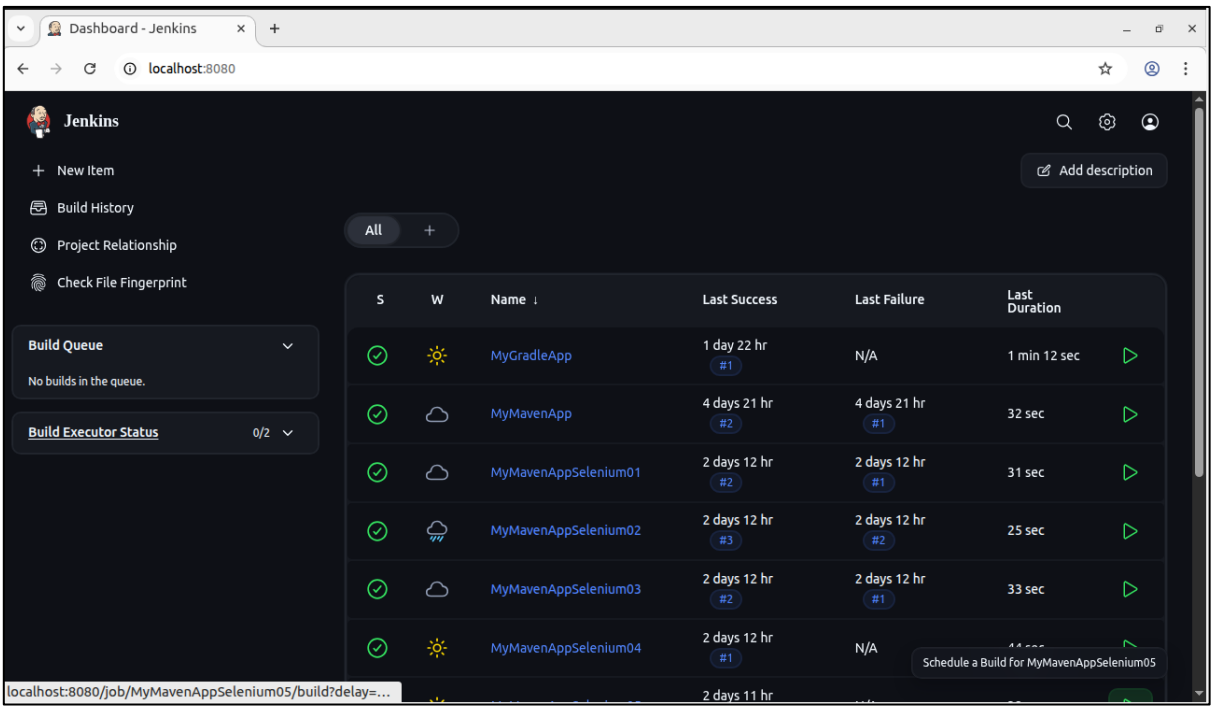




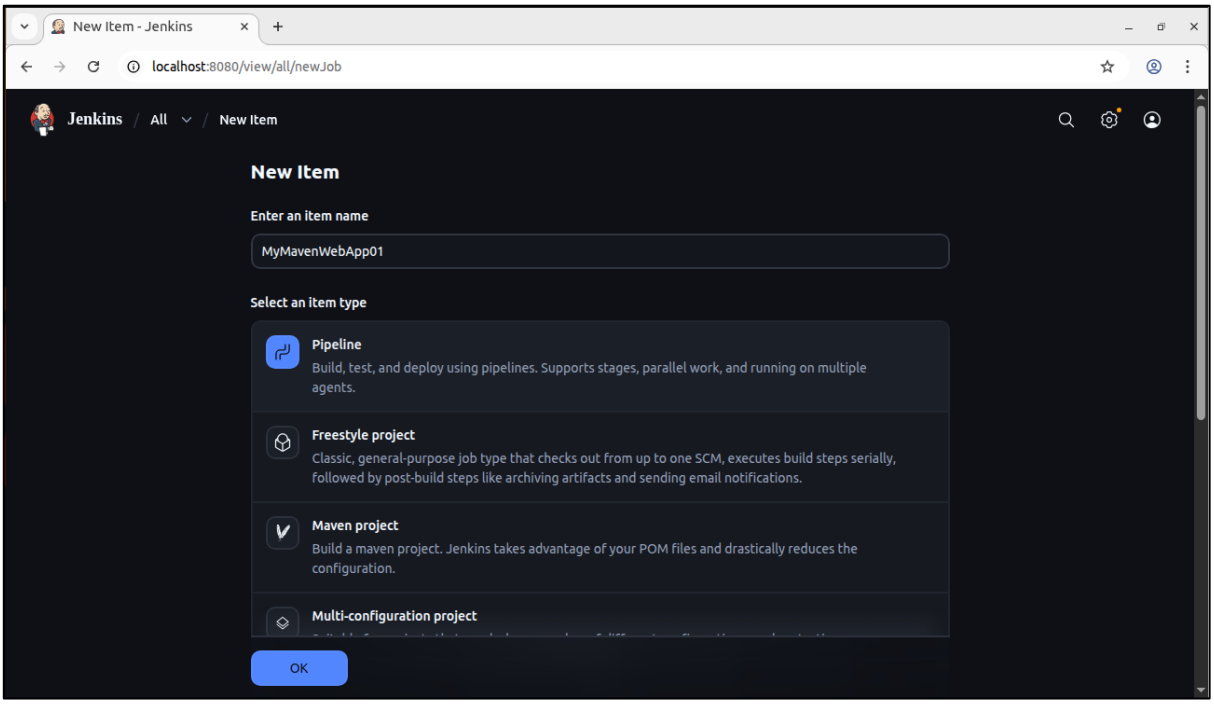
Niranjan G K "1BI23CS142"

Jenkins Pipeline for MyMavenWebApp01

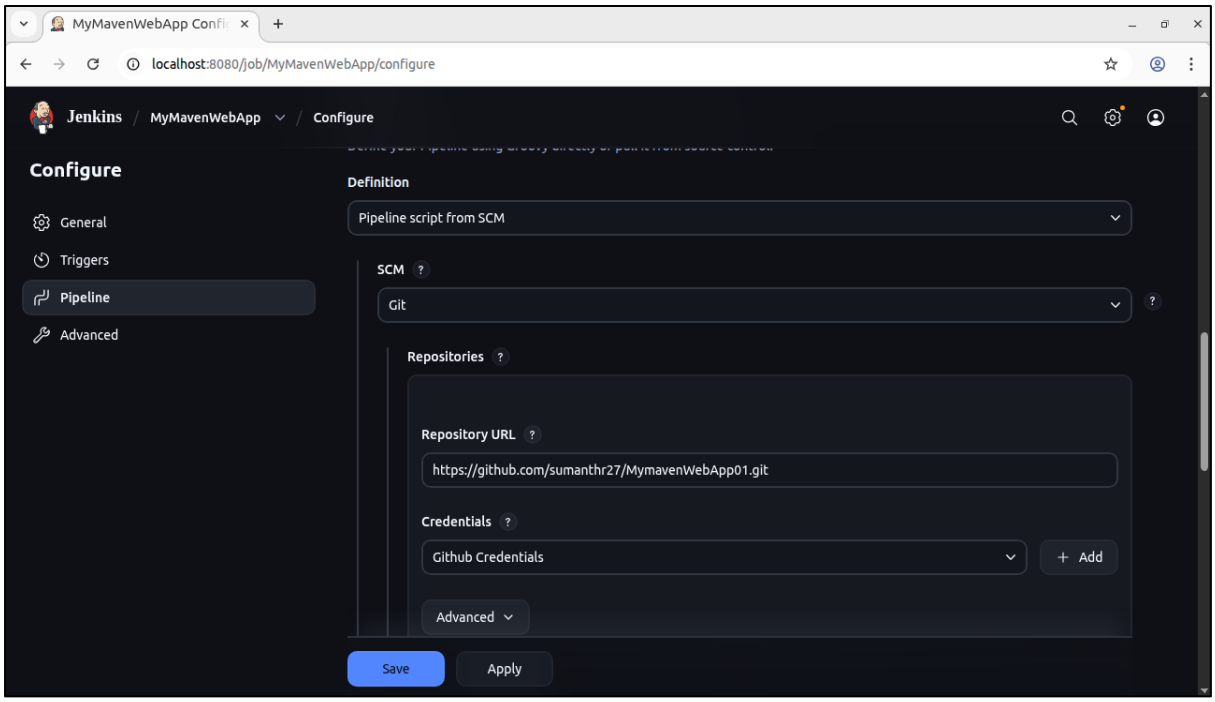
Step 1: Open the Jenkins Dashboard in order to create a new item



Step 2: Click on New Item in order for creating an New Item and thus give the name as "MyMavenWebApp



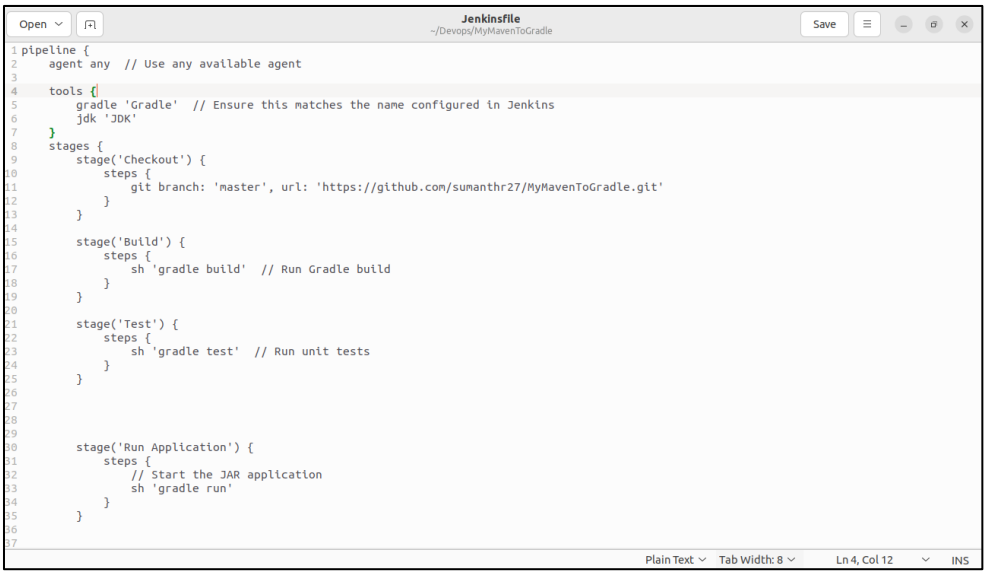
Step 3: Configure the settings by Defining the script as “Pipeline Script from SCM” and SCM as “Git”. Thus, now as we have selected Git, we need to paste the Repository URL of the project given in GitHub and choose Credentials as “GitHub Credentials” which was created earlier.



Step 4: Now in Terminal, go to directory created for MyMavenWebApp01 and create an new file called “Jenkinsfile”

```
sumanth@sumanth-VirtualBox:~$ cd Devops
sumanth@sumanth-VirtualBox:~/Devops$ cd MymavenWebApp01
sumanth@sumanth-VirtualBox:~/Devops/MymavenWebApp01$ ls
Jenkinsfile  pom.xml  src  target
sumanth@sumanth-VirtualBox:~/Devops/MymavenWebApp01$ gedit Jenkinsfle
```

Step 5: Paste the available code inside the Jenkinsfile

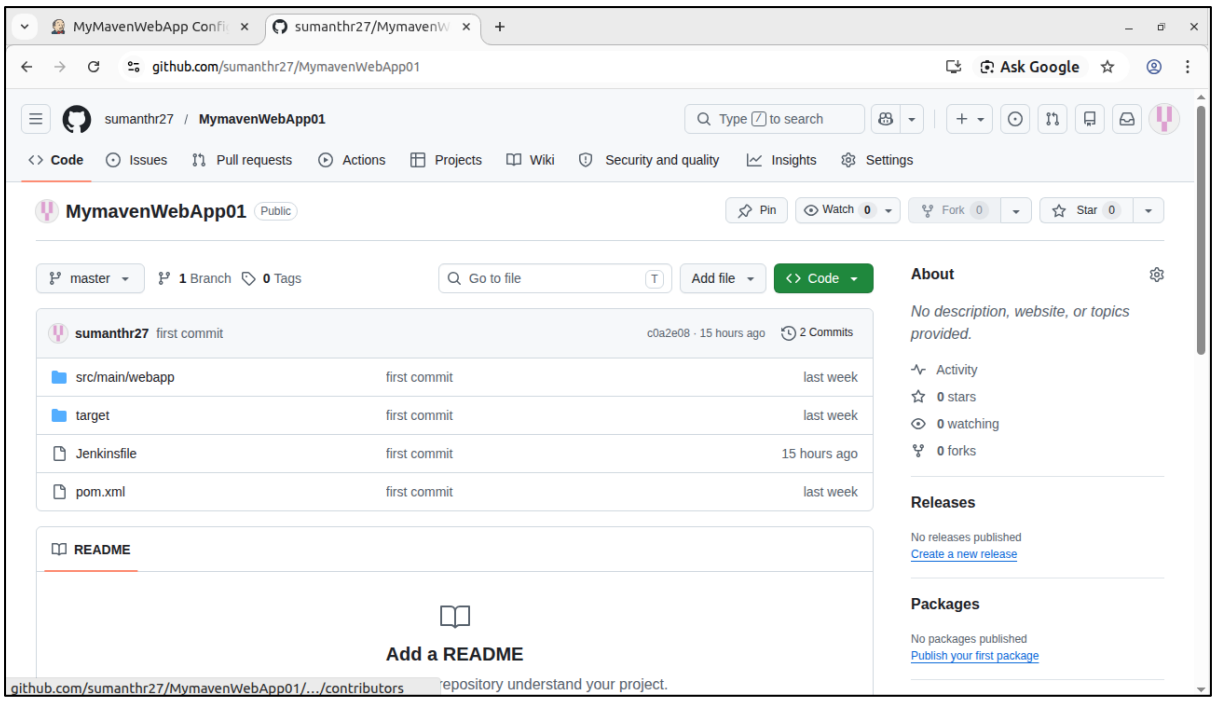



```
pipeline {
    agent any  // Use any available agent

    tools {
        gradle 'Gradle'  // Ensure this matches the name configured in Jenkins
        jdk 'JDK'
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'master', url: 'https://github.com/sumanthr27/MyMavenToGradle.git'
            }
        }

        stage('Build') {
            steps {
                sh 'gradle build'  // Run Gradle build
            }
        }

        stage('Test') {
            steps {
                sh 'gradle test'  // Run unit tests
            }
        }



        stage('Run Application') {
            steps {
                // Start the JAR application
                sh 'gradle run'
            }
        }
```

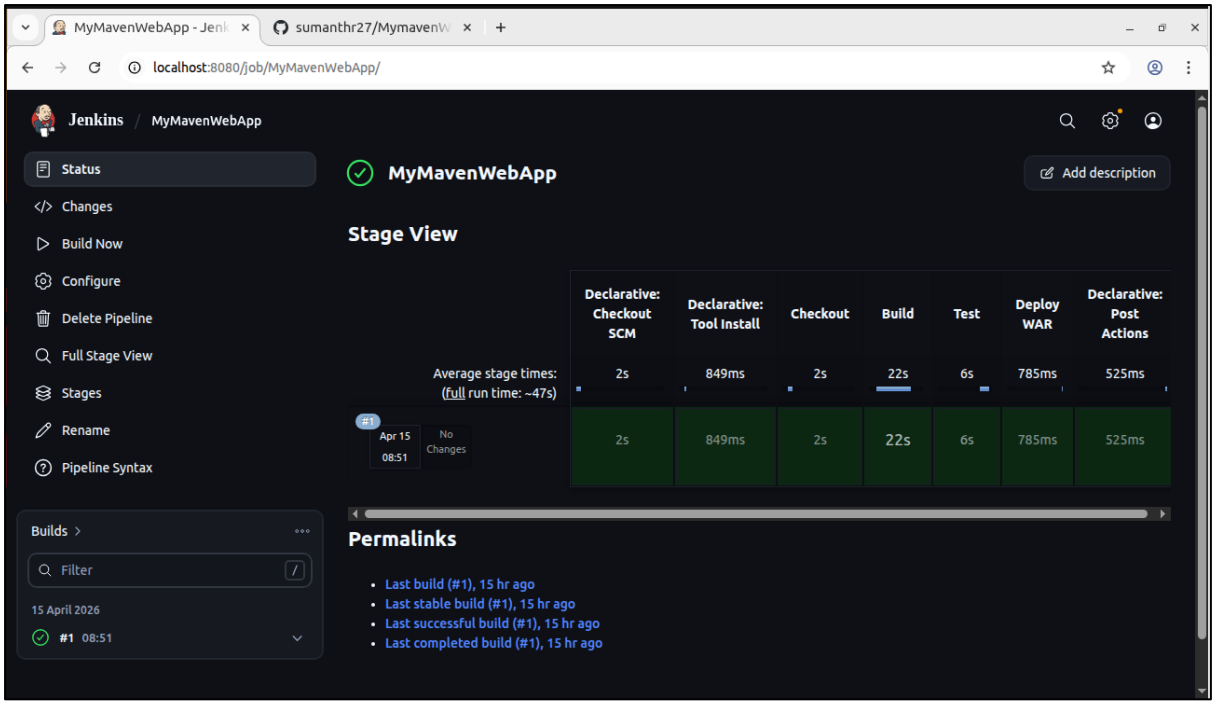
Step 6: We need to update pom.xml also in order to change the versions of source and target from 1.7 to 1.8

```
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>
```

Step 7: Now we need to Push changes and commit by pushing the necessary "Jenkinsfile" to the respective Repo

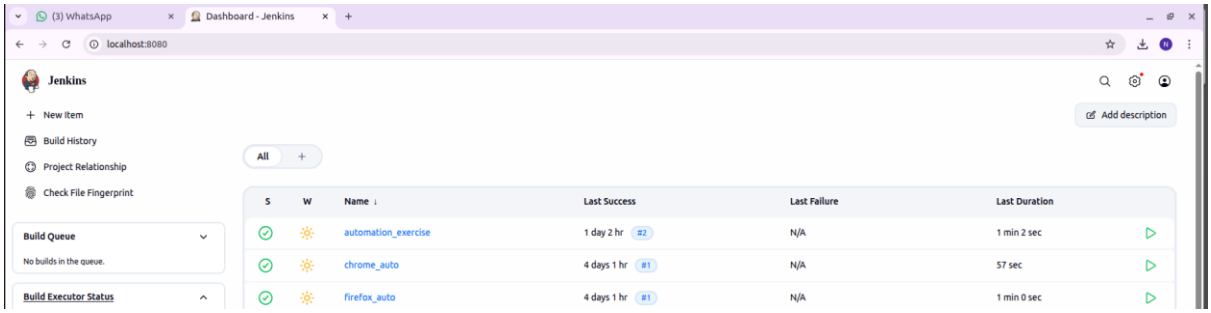


Step 8: Now go to Jenkins and Build the Pipeline and thus check the Stage view for the successful execution.

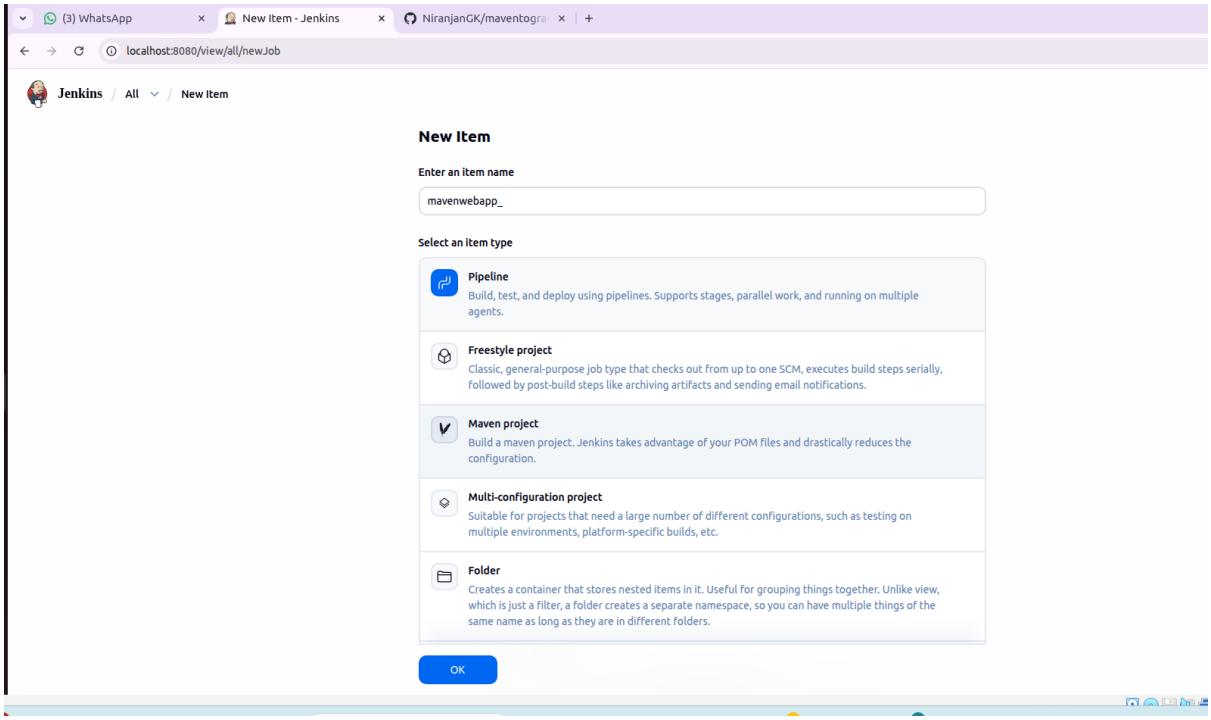

MyMavenWebApp - Jenk
sumanthr27/MymavenW
localhost:8080/job/MyMavenWebApp/
Jenkins / MyMavenWebApp
Status
Changes
Build Now
Configure
Delete Pipeline
Full Stage View
Stages
Rename
Pipeline Syntax
Builds
Filter
15 April 2026
#1 08:51
MyMavenWebApp
Add description
Stage View
Declarative: Checkout SCM
Declarative: Tool Install
Checkout
Build
Test
Deploy WAR
Declarative: Post Actions
Average stage times:
(full run time: ~47s)
2s
849ms
2s
22s
6s
785ms
525ms
#1
Apr 15
08:51
No Changes
Permalinks
Last build (#1), 15 hr ago
Last stable build (#1), 15 hr ago
Last successful build (#1), 15 hr ago
Last completed build (#1), 15 hr ago

Jenkins Pipeline for MyMavenWebApp

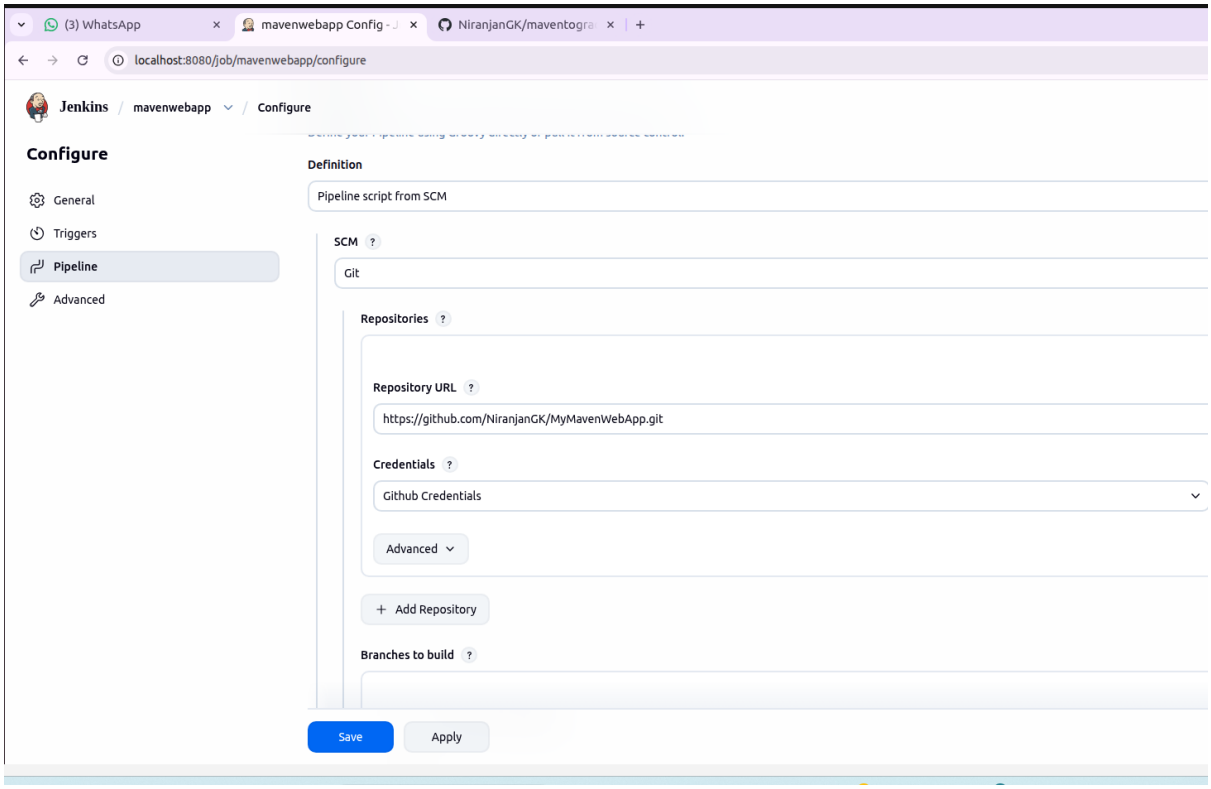
Step 1: Open the Jenkins Dashboard to create a new item



Step 2: Click on New Item in order for creating an New Item and thus give the name as "mavenwebapp" and Item type as Pipeline and click add



Step 3: Configure the settings by Defining the script as "Pipeline Script from SCM" and SCM as "Git". Thus, now as we have selected Git, we need to paste the Repository URL of the project given in GitHub and choose Credentials as "GitHub Credentials" which was created earlier.



<u>Step 4:</u> Now in Terminal, go to directory created for MyMavenAppSelenium05 and create an new file called "Jenkinsfile"

Step 5: Paste the available code inside the Jenkinsfile

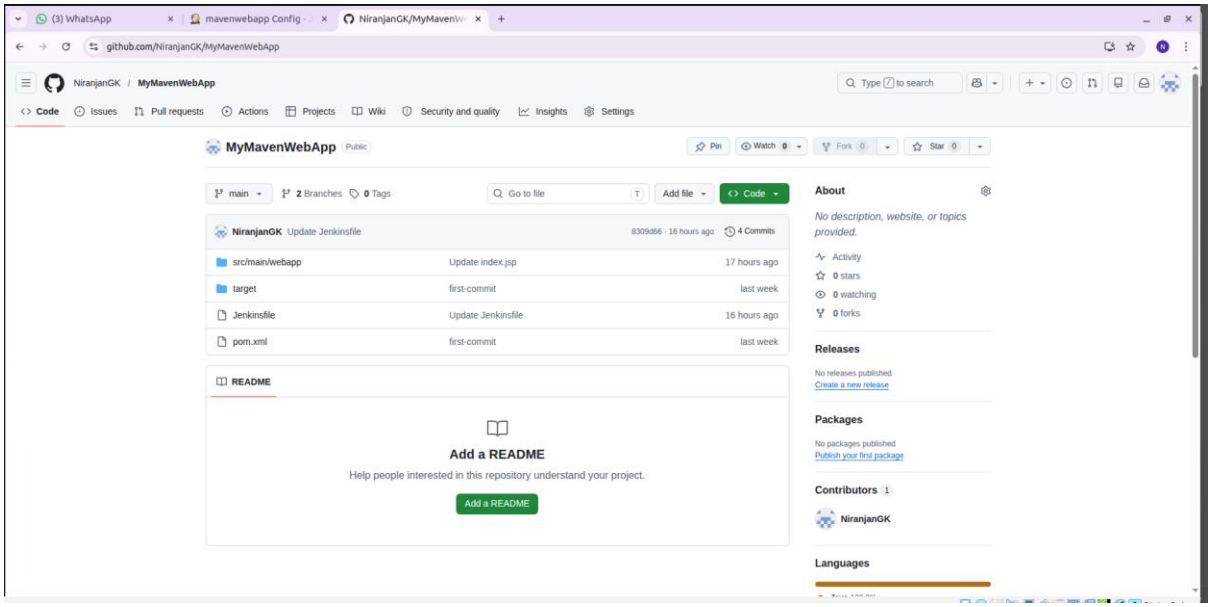
```
pipeline {
    agent any  // Use any available agent

    tools {
        gradle 'Gradle'  // Ensure this matches the name configured in Jenkins
        jdk 'JDK'
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'master', url:
'https://github.com/NiranjanGK/MyMavenWebApp.git'
            }
        }

        stage('Build') {
            steps {
                sh 'gradle build'  // Run Gradle build
            }
        }

        stage('Test') {
            steps {
                sh 'gradle test'  // Run unit tests
            }
        }
```

```
        stage('Run Application') {
            steps {
                // Start the JAR application
                sh 'gradle run'
            }
        }



    }

    post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
            echo 'Build failed!'
        }
    }
}
```

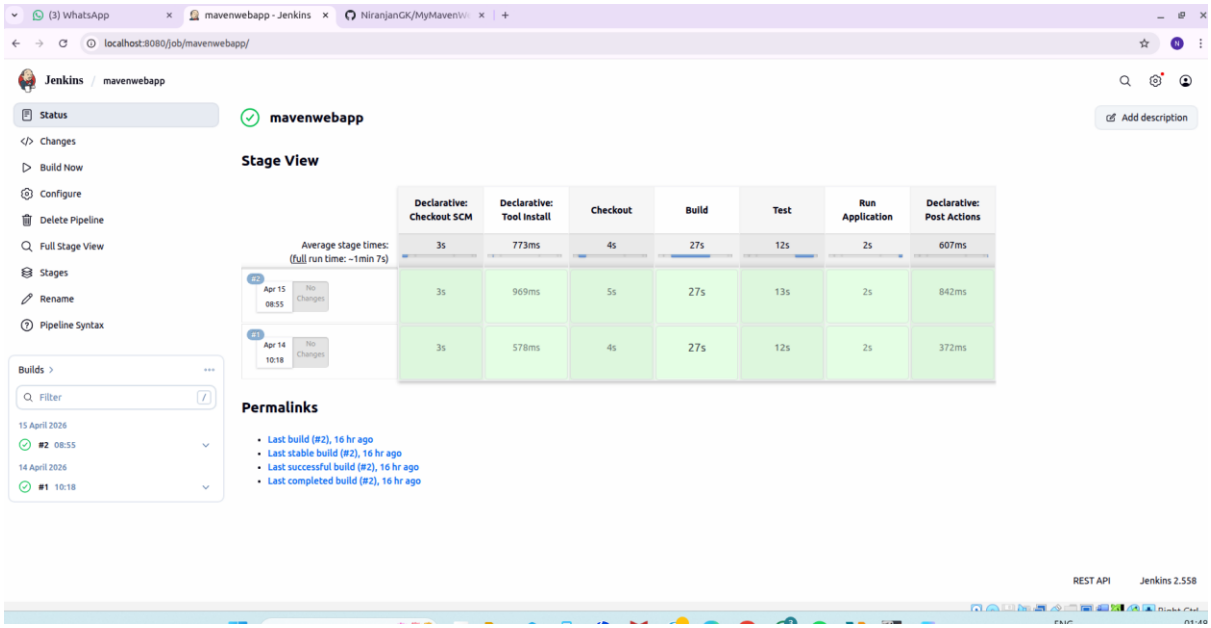
Step 6: We need to update pom.xml also in order to change the versions of source and target from 1.7 to 1.8

```
<!-- Example: Maven Compiler Plugin to compile Java code -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-compiler-plugin</artifactId>
<version>3.8.1</version>
<configuration>
<source>1.8</source>
<target>1.8</target>
</configuration>
</plugin>
```

Step 7: Now we need to Push changes and commit by pushing the necessary "Jenkinsfile" to the respective Repo



Step 8: Now go to Jenkins and Build the Pipeline and thus check the Stage view for the successful execution.


(3) WhatsApp
mavenwebapp - Jenkins
NiranjanGK/MyMavenW
localhost:8080/job/mavenwebapp/
Jenkins / mavenwebapp
Status
Changes
Build Now
Configure
Delete Pipeline
Full Stage View
Stages
Rename
Pipeline Syntax
Builds
Filter
15 April 2026
#2 08:55
14 April 2026
#1 10:18
mavenwebapp
Add description
Stage View
Declarative: Checkout SCM
Declarative: Tool Install
Checkout
Build
Test
Run Application
Declarative: Post Actions
Average stage times:
(full run time: ~1min 7s)
3s 773ms 4s 27s 12s 2s 607ms
#2 Apr 15 08:55 No Changes
3s 969ms 5s 27s 13s 2s 842ms
#1 Apr 14 10:18 No Changes
3s 578ms 4s 27s 12s 2s 372ms
Permalinks
Last build (#2), 16 hr ago
Last stable build (#2), 16 hr ago
Last successful build (#2), 16 hr ago
Last completed build (#2), 16 hr ago
REST API
Jenkins 2.558